Walchand College of Engineering, Sangli

Department of Computer Science and Engineerin

Course: High Performance Computing Lab

Practical No. 7

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Problem Statement 1:

Implement Matrix-Vector Multiplication using MPI. Use different number of processes and analyze the performance.

Program :

```c
int main(int argc, char* argv[])
{
    int rank, size;
    int m, n;
    double *A = NULL;
    double *x = NULL;
    double *y = NULL;
    double *local_A = NULL;
    double *local_y = NULL;
    int rows_per_proc;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    if (rank == 0)
    {
        printf("Enter matrix dimensions m x n (m rows, n cols): ");
        fflush(stdout);
        scanf("%d %d", &m, &n);

        A = (double*)malloc(m * n * sizeof(double));
        x = (double*)malloc(n * sizeof(double));
        y = (double*)malloc(m * sizeof(double));

        printf("Initializing matrix and vector...\n");
        for (int i = 0; i < m; i++) {
            for (int j = 0; j < n; j++) {
                A[i*n + j] = rand() % 10;
            }
        }
        for (int j = 0; j < n; j++) {
            x[j] = rand() % 10;
        }
    }

    MPI_Bcast(&m, 1, MPI_INT, 0, MPI_COMM_WORLD);
    MPI_Bcast(&n, 1, MPI_INT, 0, MPI_COMM_WORLD);

    rows_per_proc = m / size;
    int remainder = m % size;
    int local_rows = (rank < remainder) ? rows_per_proc + 1 : rows_per_proc;

```

```c
    local_A = (double*)malloc(local_rows * n * sizeof(double));
    local_y = (double*)malloc(local_rows * sizeof(double));
    if (rank != 0) x = (double*)malloc(n * sizeof(double));

    MPI_Bcast(x, n, MPI_DOUBLE, 0, MPI_COMM_WORLD);

    int *sendcounts = NULL, *displs = NULL;
    if (rank == 0) {
        sendcounts = (int*)malloc(size * sizeof(int));
        displs = (int*)malloc(size * sizeof(int));
        int offset = 0;
        for (int i = 0; i < size; i++) {
            int rows = (i < remainder) ? rows_per_proc + 1 : rows_per_proc;
            sendcounts[i] = rows * n;
            displs[i] = offset;
            offset += rows * n;
        }
    }

    MPI_Scatterv(A, sendcounts, displs, MPI_DOUBLE,
                 local_A, local_rows*n, MPI_DOUBLE,
                 0, MPI_COMM_WORLD);

    for (int i = 0; i < local_rows; i++) {
        local_y[i] = 0.0;
        for (int j = 0; j < n; j++) {
            local_y[i] += local_A[i*n + j] * x[j];
        }
    }

    if (rank == 0) {
        int *recvcounts = sendcounts;
        int *rdispls = displs;
        MPI_Gatherv(local_y, local_rows, MPI_DOUBLE,
                    y, recvcounts, rdispls, MPI_DOUBLE,
                    0, MPI_COMM_WORLD);

        printf("\nResult vector y = A * x:\n");
        for (int i = 0; i < m; i++) {
            printf("%lf ", y[i]);
        }
        printf("\n");
```

```c
    if (rank == 0) {
        int *recvcounts = sendcounts;
        int *rdispls = displs;
        MPI_Gatherv(local_y, local_rows, MPI_DOUBLE,
                    y, recvcounts, rdispls, MPI_DOUBLE,
                    0, MPI_COMM_WORLD);

        printf("\nResult vector y = A * x:\n");
        for (int i = 0; i < m; i++) {
            printf("%lf ", y[i]);
        }
        printf("\n");

        free(A); free(x); free(y);
        free(sendcounts); free(displs);
    } else {
        MPI_Gatherv(local_y, local_rows, MPI_DOUBLE,
                    NULL, NULL, NULL, MPI_DOUBLE,
                    0, MPI_COMM_WORLD);
        free(x);
    }

    free(local_A);
    free(local_y);

    MPI_Finalize();
    return 0;
}
```

Output :

```
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment7$ mpicc ps1.c -o ps1
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment7$ mpirun --oversubscribe -np 5 ./ps1
Enter matrix dimensions m x n (m rows, n cols): 3
3
Initializing matrix and vector...

Result vector y = A * x:
64.000000 0.000000 0.000000
double free or corruption (out)
[Vivobook16x-Soham:00547] *** Process received signal ***
[Vivobook16x-Soham:00547] Signal: Aborted (6)
[Vivobook16x-Soham:00547] Signal code:  (-6)
[Vivobook16x-Soham:00547] [ 0] /lib/x86_64-linux-gnu/libc.so.6(+0x45330)[0x7f83e7981330]
[Vivobook16x-Soham:00547] [ 1] /lib/x86_64-linux-gnu/libc.so.6(pthread_kill+0x11c)[0x7f83e79dab2c]
[Vivobook16x-Soham:00547] [ 2] /lib/x86_64-linux-gnu/libc.so.6(gsignal+0x1e)[0x7f83e798127e]
[Vivobook16x-Soham:00547] [ 3] /lib/x86_64-linux-gnu/libc.so.6(abort+0xdf)[0x7f83e79648ff]
[Vivobook16x-Soham:00547] [ 4] /lib/x86_64-linux-gnu/libc.so.6(+0x297b6)[0x7f83e79657b6]
[Vivobook16x-Soham:00547] [ 5] /lib/x86_64-linux-gnu/libc.so.6(+0xa8ff5)[0x7f83e79e4ff5]
[Vivobook16x-Soham:00547] [ 6] /lib/x86_64-linux-gnu/libc.so.6(+0xab120)[0x7f83e79e7120]
[Vivobook16x-Soham:00547] [ 7] /lib/x86_64-linux-gnu/libc.so.6(+0xab43a)[0x7f83e79e743a]
[Vivobook16x-Soham:00547] [ 8] /lib/x86_64-linux-gnu/libc.so.6(__libc_free+0x7e)[0x7f83e79e9dae]
[Vivobook16x-Soham:00547] [ 9] /lib/x86_64-linux-gnu/libmpi.so.40(+0xa71a9)[0x7f83e7bf51a9]
[Vivobook16x-Soham:00547] [10] /lib/x86_64-linux-gnu/libmpi.so.40(+0xa2039)[0x7f83e7bf0039]
[Vivobook16x-Soham:00547] [11] /lib/x86_64-linux-gnu/libmpi.so.40(mca_coll_base_comm_unselect+0x1f21)[0x7f83e7bf21c1]
[Vivobook16x-Soham:00547] [12] /lib/x86_64-linux-gnu/libmpi.so.40(+0x30217)[0x7f83e7b7e217]
[Vivobook16x-Soham:00547] [13] /lib/x86_64-linux-gnu/libmpi.so.40(ompi_comm_finalize+0xf9)[0x7f83e7b7dee9]
[Vivobook16x-Soham:00547] [14] /lib/x86_64-linux-gnu/libmpi.so.40(ompi_mpi_finalize+0x61d)[0x7f83e7bb057d]
[Vivobook16x-Soham:00547] [15] ./ps1(+0x19c2)[0x55a091d719c2]
[Vivobook16x-Soham:00547] [16] /lib/x86_64-linux-gnu/libc.so.6(+0x2a1ca)[0x7f83e79661ca]
[Vivobook16x-Soham:00547] [17] /lib/x86_64-linux-gnu/libc.so.6(__libc_start_main+0x8b)[0x7f83e796628b]
[Vivobook16x-Soham:00547] [18] ./ps1(+0x1265)[0x55a091d71265]
[Vivobook16x-Soham:00547] *** End of error message ***
--------------------------------------------------------------------------
Primary job  terminated normally, but 1 process returned
a non-zero exit code. Per user-direction, the job has been aborted.
--------------------------------------------------------------------------
--------------------------------------------------------------------------
mpirun noticed that process rank 0 with PID 0 on node Vivobook16x-Soham exited on signal 6 (Aborted).
--------------------------------------------------------------------------
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment7$
```

Problem Statement 2:

Implement Matrix-Matrix Multiplication using MPI. Use different number of processes and analyze the performance.

Screenshot:

```
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment7$ mpicc ps2.c -o ps2
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment7$ mpirun --oversubscribe -np 5 ./ps2
Enter dimensions m n p for A(m×n) * B(n×p): 3
3
4
Initializing matrices A and B...
[Vivobook16x-Soham:00572] *** An error occurred in MPI_Gatherv
[Vivobook16x-Soham:00572] *** reported by process [544342017,0]
[Vivobook16x-Soham:00572] *** on communicator MPI_COMM_WORLD
[Vivobook16x-Soham:00572] *** MPI_ERR_TRUNCATE: message truncated
[Vivobook16x-Soham:00572] *** MPI_ERRORS_ARE_FATAL (processes in this communicator will now abort,
[Vivobook16x-Soham:00572] ***    and potentially your MPI job)
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment7$
```

GitHub Link : **HPCL/Assignment7 at main · Somzee5/HPCL**